Apr 28 2023 | Logged in as : Goytia, Renzo Nicolas NicolÃ¡s

Introduction

Baroid Core Homework 16

Assessment Feedback

Congratulations, you passed.

Total score: 56 out of 64, 88%

Question Feedback

Question

1 of 16

In order to change the NAP/Water ratio from 70/30 to 85/15, we need to add water.

- ○ True
- ◉ False

1 out of 1

Question

2 of 16

When you change the NAP/Water ratio, will the fluid density increase?

- ○ Yes
- ◉ No

1 out of 1

Question

3 of 16

In order to increase the WPS of an IEF fluid, which value(s) do we need know? (select all applicable options)

- [x] Desired concentration of salt
- [ ] Current NAP/Water ratio
- [x] Current water phase salinity

☐ Density of the base fluid

2 out of 2

Question

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Calculate the oil, water, calcium chloride (80lb/sack), and barite (100lb/sack) required to mix 200 bbl of a 15.5 ppg invert emulsion drilling fluid with a WPS of 220,000ppm and NAP/water ratio of 75/25. The density of the base oil is 7.0ppg.

How many barrels of base fluid are needed? = 107.38

3 out of 3

Question

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For Non aqueous fluid build 1, how many barrels of water are required?

Barrels of water = 33.22

3 out of 3

Question

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For Non aqueous fluid build 1, how many sacks of barite(100lb/sack) are needed?

Sacks of barite(100lb/sack) = 837

3 out of 3

Question

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For Non aqueous fluid build 1, how many sacks of calcium chloride (80lb/sack) are needed?

Sacks of calcium chloride (80lb/sack) = 43

3 out of 3

Question

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For OWR adjustment 1, how many barrels of non aqueous phase (base oil) are needed?

Barrels of NAP = 560

3 out of 3

Question

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For OWR adjustment 1, how many barrels of water are needed?

Barrels of water = 0

3 out of 3

Question

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For OWR adjustment 1, how many sacks of barite (100lb/sack) are needed?

Sacks of barite (100lb/sack) = 3120

3 out of 3

Question

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For OWR adjustment 2, how many barrels of NAP are needed?

Barrels of NAP = 0

1 out of 1

Question

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For OWR adjustment 2, how many barrels of water are needed?

Barrels of water = 192

3 out of 3

Question

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For OWR adjustment 2, how many sacks of barite (100lb/sack) are required?

Sacks of barite (100lb/sack) = 420

3 out of 3

Question

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Given the following data for 1000 bbl of a 10.5ppg IEF drilling fluid:
NAP: 67% Water: 20% Solids: 13%
WPS: 230,000ppm

Calculate how much 97% pure $CaCl_2$ (80lb/sack) must be added to adjust the WPS to 280,000ppm.

Sacks of calcium chloride (80lb/sack) = 82

3 out of 3

Question

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Given the following data for 1500bbl of a 14.0ppg IEF drilling fluid:
NAP: 63% Water: 15% Solids: 22%
WPS: 270,000ppm.

Calculate how much water must be added to adjust the WPS to 220,000ppm.

Barrels of water = 152.36

0 out of 3
Water = 63.14bbl

Question

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Results of the WPS Analysis

What is the NAP ratio in %? (whole number) = 81 What is the Water ratio in % (whole number) = 19 Water Phase Salinity in ppm = 278100 Specific gravity of the brine (5 decimals) = 1.25557 %Volume fraction of the brine = 14.34 ASG (3 decimals) = 3.774 %LGS = 6.3 LGSppb = 57.41 %HGS = 17.36 HGSppb = 255.56

21 out of 26

NAP/Water ratio	83/17 %

WPS	278100 ppm

Specific Gravity of brine	1.25554 sg

% Volume fraction of the brine	14.34 %

ASG	3.774 sg

Low gravity Solids	6.3 %	57.41 ppb

High gravity Solids	17.36 %	255.56 ppb